

HEALTHCARE DATA CLIENT

KPI Catalogue

Healthcare Data Quality Automation and Exception Reporting

Document Control	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Batch Scale	Approximately 250,000 records per batch
Confirmed Quality Result	2% exception rate
Confirmed Turnaround Result	6 hours to 2.5 hours per file
Catalogue purpose: Defines the governed measures used to describe validation volume, exceptions and review turnaround without inventing unsupported healthcare metrics.	

1. KPI Summary

KPI ID	KPI / Measure	Definition	Evidence Status
KPI-01	Records Processed	Total records included in the processed batch.	Confirmed analytical measure
KPI-02	Exception Records	Distinct processed records failing one or more applicable validation rules.	Confirmed analytical measure
KPI-03	Exception Rate	Exception Records / Records Processed x 100.	Confirmed rule and result
KPI-04	Non-Exception Records	Processed records that do not fail an applicable validation rule.	Reconstructed supporting measure
KPI-05	Validation Failures	Total rule-level failures generated across processed records.	Reconstructed supporting measure
KPI-06	Exceptions by Reason Code	Exception records grouped by validation reason code or failure explanation.	Confirmed analytical capability
KPI-07	Review Turnaround Time	Elapsed review time per file under the applicable review process.	Confirmed outcome measure
KPI-08	Review Time Reduction	Baseline Review Time minus Post-Solution Review Time.	Derived outcome measure
KPI-09	Review Turnaround Reduction %	Review Time Reduction / Baseline Review Time x 100.	Derived outcome measure

2. KPI-01 Records Processed

Attribute	Definition
Purpose	Establish the denominator and processing-volume context for each batch.
Calculation	Count of records included in the processed batch.
Confirmed scale	Approximately 250,000 records per batch.
Validation	Reconcile processed population to the supported input after documented ingestion controls.
Boundary	The approximate 250,000 figure is batch scale, not a claim that every batch contained exactly 250,000 records.

3. KPI-02 Exception Records

Attribute	Definition
Purpose	Measure the number of records requiring review after validation.
Rule	A record failing one or more applicable validation rules is treated as an exception.
Counting principle	A record should be counted once in the exception-record population even if it fails multiple rules.
Output	Exception file, table or report with reason code or equivalent failure explanation.
Boundary	Exact historical exception-record counts are not retained.

4. KPI-03 Exception Rate

Attribute	Definition
Formula	Records failing one or more validation rules / Total records processed x 100
Observed result	2%
Numerator	Exception Records
Denominator	Records Processed
Interpretation	Share of processed records requiring review because at least one validation rule failed.
Boundary	2% is an observed engagement result, not a target, SLA, tolerance or acceptable-error threshold.

Critical distinction: Exception Rate is record-based. A record with multiple validation failures remains one exception record in the numerator.

5. Supporting Data Quality Measures

KPI	Definition	Implementation Note
Non-Exception Records	Records Processed minus Exception Records.	Portfolio supporting measure. It must not be described as a historical client KPI unless evidence is available.
Validation Failures	Count of individual rule failures.	One record may contribute multiple failures. Therefore this measure can exceed Exception Records.
Exceptions by Reason Code	Exception records grouped by reason code or failure explanation.	Useful for identifying which validation categories contribute to review workload. Exact historical category counts are not retained.

6. Validation Dimensions

Validation Dimension	Confirmed Scope
Duplicates	Duplicate-record checks
Completeness	Null and missing-value checks
Dates	Invalid-date checks
Formats	Incorrect-format checks
Identifiers	Invalid-ID checks
Ranges	Out-of-range checks
Codes	Inconsistent-code checks
Cross-field integrity	Cross-field mismatch checks

Rule boundary: Exact field-level thresholds, valid code lists, identifier formats and cross-field conditions are not retained. They must not be fabricated in the portfolio.

7. KPI-07 Review Turnaround Time

Attribute	Definition
Purpose	Measure the elapsed time required to review a file.
Baseline	6 hours per file
Post-solution	2.5 hours per file
Unit	Hours per file
Confirmed change	3.5 fewer hours of review turnaround per file
Boundary	The exact detailed manual steps within the original 6-hour process are not retained.

8. KPI-08 Review Time Reduction

Attribute	Definition
Formula	Baseline Review Time - Post-Solution Review Time
Calculation	6.0 hours - 2.5 hours
Result	3.5 hours per file
Status	Derived directly from confirmed values

9. KPI-09 Review Turnaround Reduction %

Attribute	Definition
Formula	(Baseline Review Time - Post-Solution Review Time) / Baseline Review Time x 100
Calculation	(6.0 - 2.5) / 6.0 x 100
Result	Approximately 58.3%
Status	Derived measure
Claim boundary	Describes turnaround reduction only. It is not a productivity, labor-cost, clinical-quality or compliance metric.

10. KPI Relationships

Relationship	Control
Records Processed to Exception Records	Exception Records must be a subset of the processed population.
Exception Records to Exception Rate	Exception Records form the numerator; Records Processed form the denominator.
Exception Records to Validation Failures	One exception record may have one or multiple validation failures.
Reason Codes to Exceptions	Reason codes explain validation failures and support exception analysis.
Baseline to Post-Solution Review Time	Both values use the same unit: hours per file.
Review Time Reduction to Reduction %	Percentage is derived from the confirmed baseline and post-solution values.

11. KPI Validation Controls

Control	Expected Result
Population reconciliation	Records Processed reconciles to the governed batch population.
Exception reconciliation	Each Exception Record fails at least one applicable validation rule.
No double-counting	A record failing multiple rules is counted once in Exception Records.
Failure traceability	Rule-level failures remain traceable to the affected record and reason.
Rate reconciliation	Exception Rate recalculates correctly from numerator and denominator.
Turnaround calculation	6 hours to 2.5 hours produces a 3.5-hour reduction and approximately 58.3% reduction.

12. Requirements Traceability

KPI / Measure	BRD Trace	Downstream Use
Records Processed	BR-01, BR-02	Validation workflow; reporting
Exception Records	BR-10, BR-11	Exception analysis; testing
Exception Rate	BR-14; RULE-01	Power BI reporting; findings
Exceptions by Reason Code	BR-12, BR-13	Exception analysis; dashboard
Review Turnaround Time	Business outcome	Final report; findings
Review Time Reduction	Derived from confirmed outcome	Final report
Review Turnaround Reduction %	Derived from confirmed outcome	Final report

13. Catalogue Integrity

Use in portfolio: The confirmed headline measures are approximately 250,000 records per batch, a 2% observed exception rate, and review turnaround reduced from 6 hours to 2.5 hours per file. Supporting measures are included only where they can be derived or implemented without inventing historical healthcare results.